

Systems Analysis of Power and Water Cogeneration: A Case Study in Accra, Ghana



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Thesis Structure

Chapter 1 - Introduction

Chapter 2- System Optimization

Chapter 3- System Features

Chapter 4 - System Cost and Organization

Chapter 5 - Affordability and Access

Chapter 6 - Conclusion

Systems Analysis of Urban Infrastructure

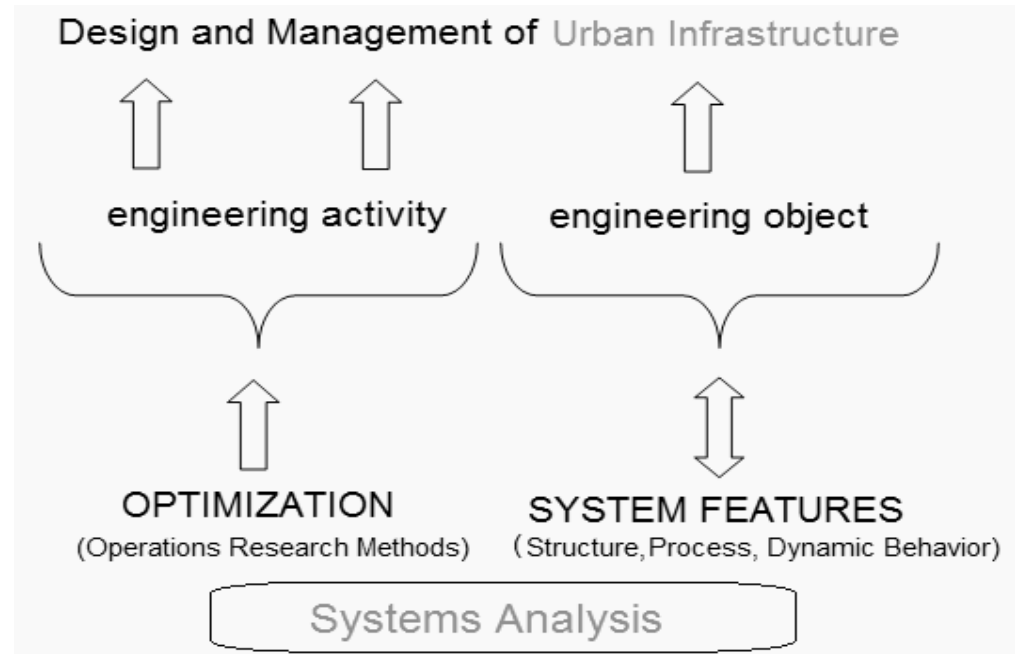


Image source: (Arnold, 2010)

Abstract

- The urban area of Accra, Ghana faces a number of imminent challenges related to meeting infrastructure demands. The issue of **water supply for a growing population** is of particular concern.
- Given the **water-energy nexus**, the question of water supply cannot be addressed without also considering energy infrastructure needed to treat, pump, and transport water.
- This study takes a systems analysis approach to the **design and management of water and energy infrastructure in the Accra urban area**. Optimization methods are used to determine the type of engineering activity required to meet forecasted demand, while system features are analyzed to determine details of scale, process, and cost.
- The identified solution (cogeneration of water and power using seawater desalination) is then analyzed in a broader **water policy** context to determine the degree to which it would meet assessed needs.



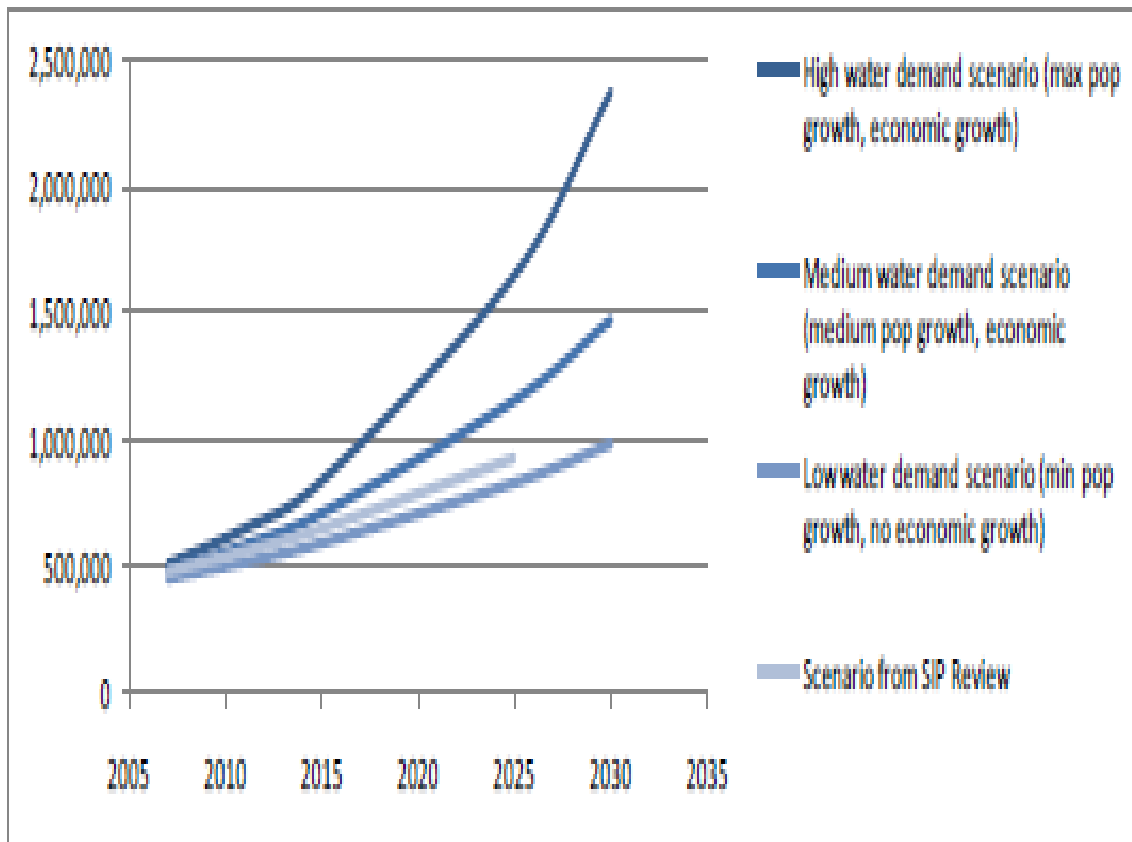
Background:

Ghana Water Company Limited is planning new supply and treatment facilities near Accra, Ghana. This includes a 60,000m³/day desalination facility which will be the first large-scale desalination contract ever to be signed in West Africa, raising a number of unique technical and institutional questions:



- **Is desalination needed** in a country such as Ghana, with access to freshwater resources?
- **Which desalination method** best suits circumstances and Best Available Technology (BAT)?
- **Is there sufficient energy** for the water supply facility; which type of power facility would be most appropriate?
- **Is desalination the best alternative** to meet human needs for water?
- **Environmental Impact Assessment (EIA)** law requires project justification and assessment of the possible alternatives to the initiative.

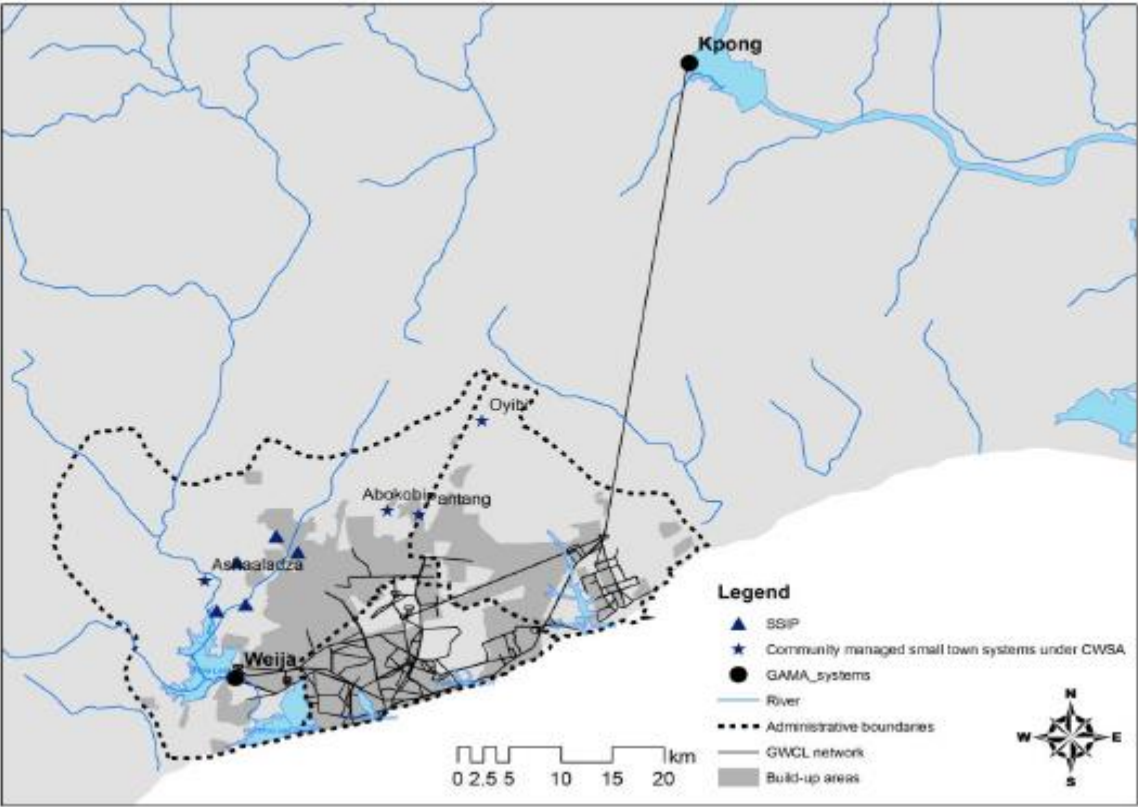
Water Supply Demand



- Currently, the utility provider GWCL (Ghana Water Company Limited) supplies 365,000m³/day,
- Current demand estimated to be up to 500,000m³/day (UN, 2010)

Projected water demand scenarios for the Greater Accra Region.

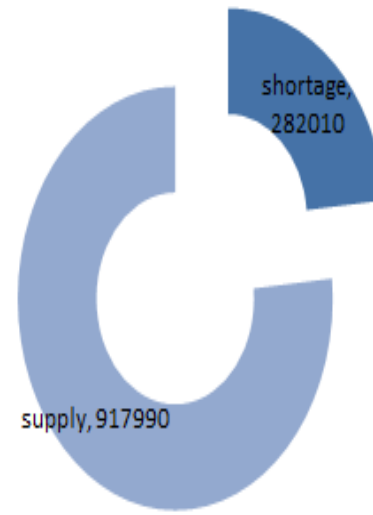
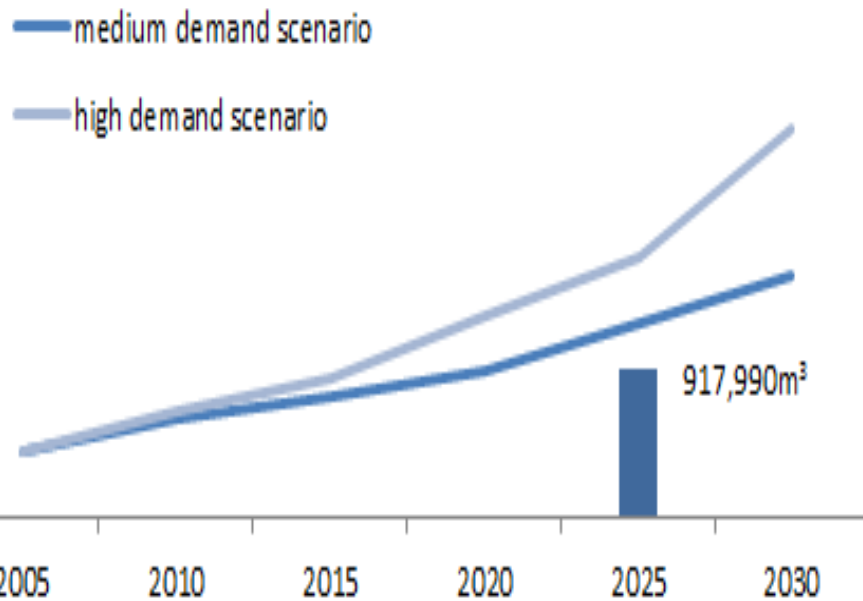
Current System Capacity and Planned Expansions



System	Capacity (m ³ /day)
Existing (Weija and Volta)	386,362
Weija expansion	68,190
new WTP from Volta	300,000
new Volta works	127,038
Total with planned expansions	881,590
Volta old	36,400
Total	917,990

The two primary water supply sources in the GAMA – the Kpong dam on the Volta River to the Northeast and the Weija Lake on the Densu River to the West of the city.

Forecasted Water Demand vs. Supply



Moderate demand scenario



High demand scenario



Own diagrams adapted from SWITCH EU, 2011 data.

Selection of Optimal Water Supply

Source	Capacity	Comments
Weija	273,000m ³ /day	Current withdrawals of 170,000m ³ /day are approaching capacity; Pollution is concern.
Volta	90,400,000m ³ /day	Plenty of supply; Transport over 50km, leads to higher costs; overreliance on one source raises water security concerns.
Groundwater	100,000m ³ /day	Brackish groundwater would require desalination. Large groundwater withdrawals generally not possible.
Rainwater Harvesting	55,000 m ³ /day	Not suitable on large scale; treatment issues if used as drinking water.
Wastewater	500,000 m ³ /day	Current WWT and storm water infrastructure is inadequate. Waste water withdrawal from coastal lagoons is brackish.



Source: adapted from SWITCH EU, 2011

Multi-Criteria Analysis

- i. **Potential:** determination of the water supply potential, based on volume relative to demand.
- ii. **Economic costs:** costs for each scenario, considering construction, transport costs and operation and maintenance O&M costs.
- iii. **Water quality:** quality of source water regarding current and potential contamination.
- iv. **Integrated Water Management:** extent to which the water supply may integrate into a waste water management, conservation, and flood management plan for the region.
- vi. **Security:** vulnerability of supply source to natural disaster, drought, security threat, infrastructure failure, etc.

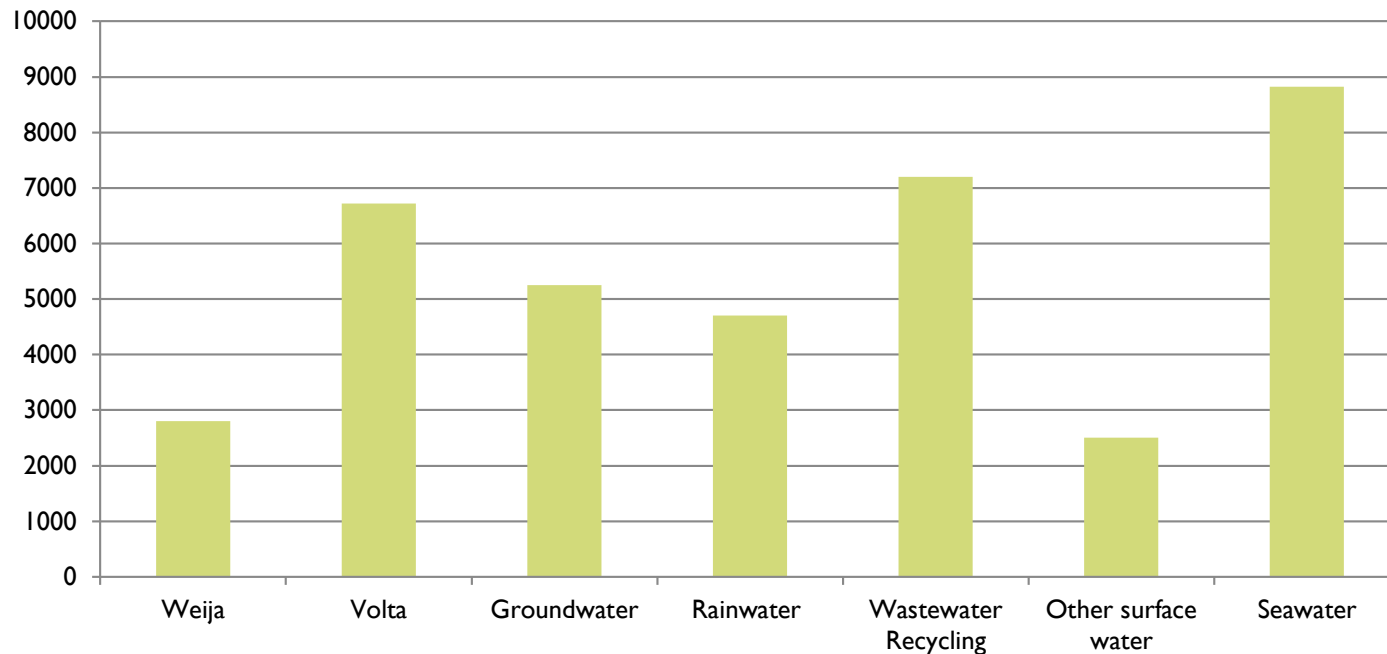
$$A_j^{water} = \prod w_j AC_j$$

Weight	Description
8-9	Great potential: low cost, high quality, integrated into regional water management, high security.
5-7	Intermediate potential; economically/technically feasible but sub-optimal.
1-4	Low potential, high cost, low quality, etc.
0	Option does not exist, not technically feasible, economically prohibitive, etc.



Source: adapted from Rujula & Dia, 2010

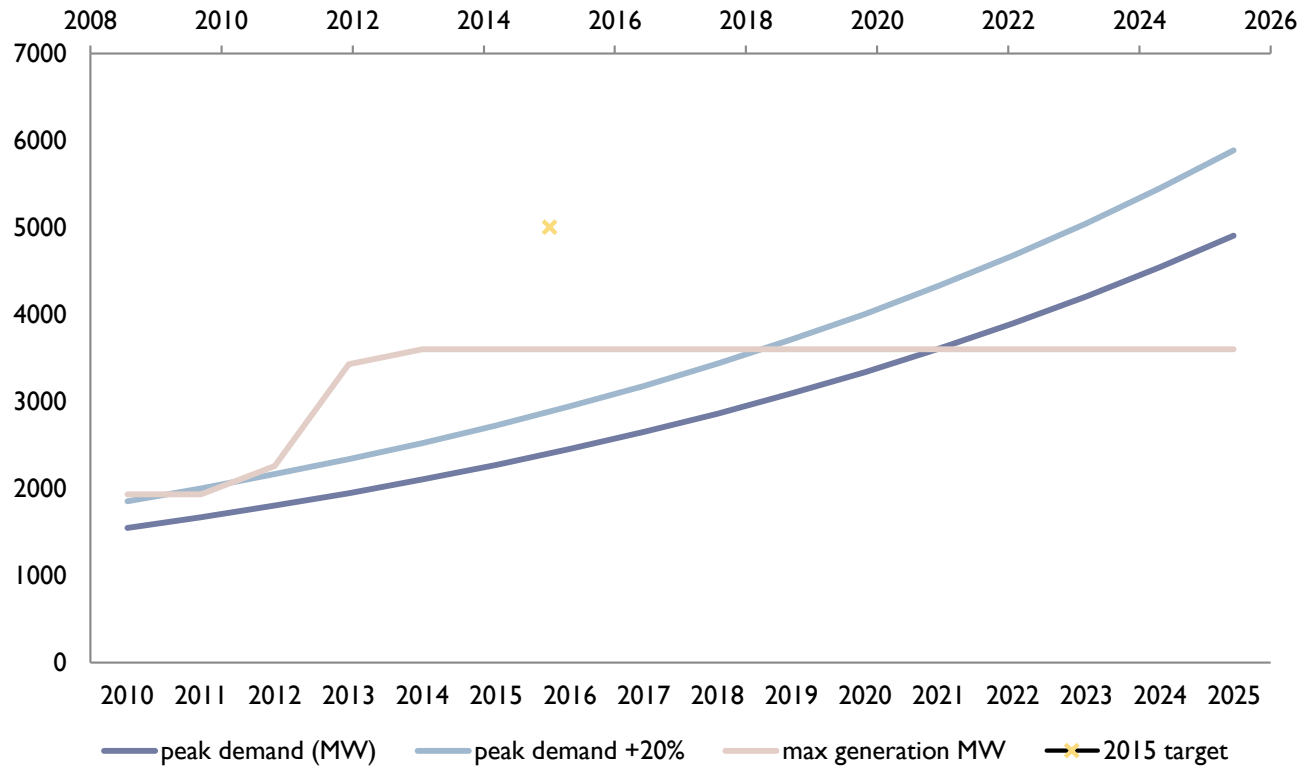
Weighted Product Model



Seawater is judged the optimal supply solution due to its abundant volume, low risk of contamination, and the fact that it diversifies existing sources providing security of supply.

Lake Volta is likely the optimal supply *ceteris paribus*, but in this particular time and context, considering water security concerns and heavy reliance on Kpong sources, desalination was given a preferential rating.

Energy Resource Assessment



Significant generation capacity must be added within the next decade in order to meet domestic demand.

Operating Plants

Plant	Type	Max. Capacity (MW)
Akosombo	Hydro	1020.0
Kpong Hydroelectric	Hydro	152
Takorandi Thermal T1	NG/Diesel	330
Takorandi Thermal T2	NG/Diesel	220
Tema Power Plant T1	NG/Diesel	113
Mines Reserve Plant	Diesel	50
Tema Power Plant T2	NG/Diesel	50
Sunon-Asogli Plant	NG	200
Osagyefo Barge	NG/Diesel	125
TOTAL		2260MW

Recent Expansions

Plant	Type	Max. Capacity (MW)	Expected completion
Kpone Power Plant	Thermal	320	2012
Tema Osono	Thermal	120	2012
Takorandi T2,T3	Thermal	220	2012
Bui Hydroelectric	Hydroelectric	400	2013
CENPOWER	Thermal	110	2013
Total Additions		1170MW	
TOTAL EXISTING AND NEW		3430MW	

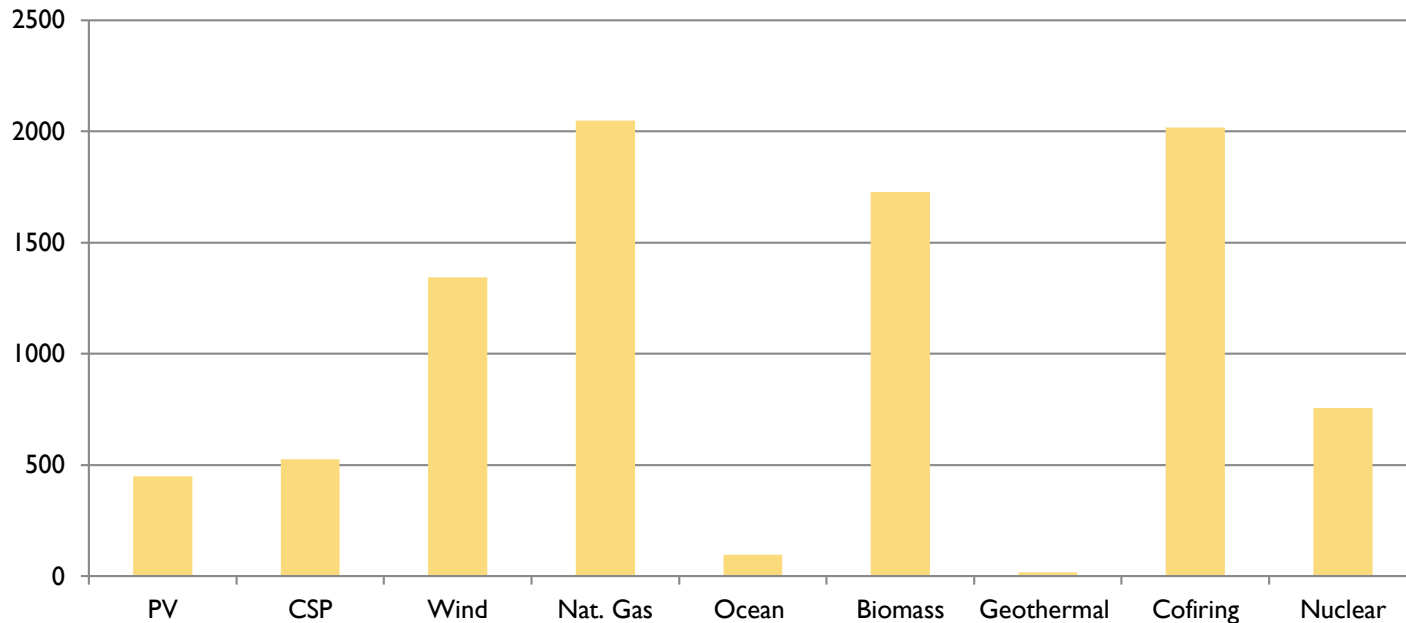


Energy Resource Assessment

Energy Resource	Assessment
Photovoltaic PV	Very large land area is required to provide sufficient energy for this scale of desalination.
Concentrated Solar Power CSP	Solar radiation resource is not ideal in this region compared to North Africa or Middle East.
Wind	Lower end of wind energy classification, makes wind energy a possibility but not ideal.
Geothermal	No known geothermal resources near site.
Fossil Fuels	Recently discovered oil and gas reserves make fossil fuels an important future source of energy in Ghana, with the prospect of the country becoming an energy exporter; significant policy commitments to natural gas have been made; environmental concerns with emissions and climate change.
Ocean	The ocean energy resources are not sufficient. Salinity gradient power is an interesting prospect for the future.
Nuclear	No current facilities but plans for the future.
Biomass	Estimated to be sufficient amounts of biomass available in Ghana for significant bio based energy expansion (based on 2011 study); collection would be a problem due to poor roads and lack of organized waste systems.



Multi-Criteria Analysis: Energy



Natural Gas is the optimal. This conclusion is due to its large potential, preferable economics, and practicality given policy commitment to natural gas infrastructure in the region.

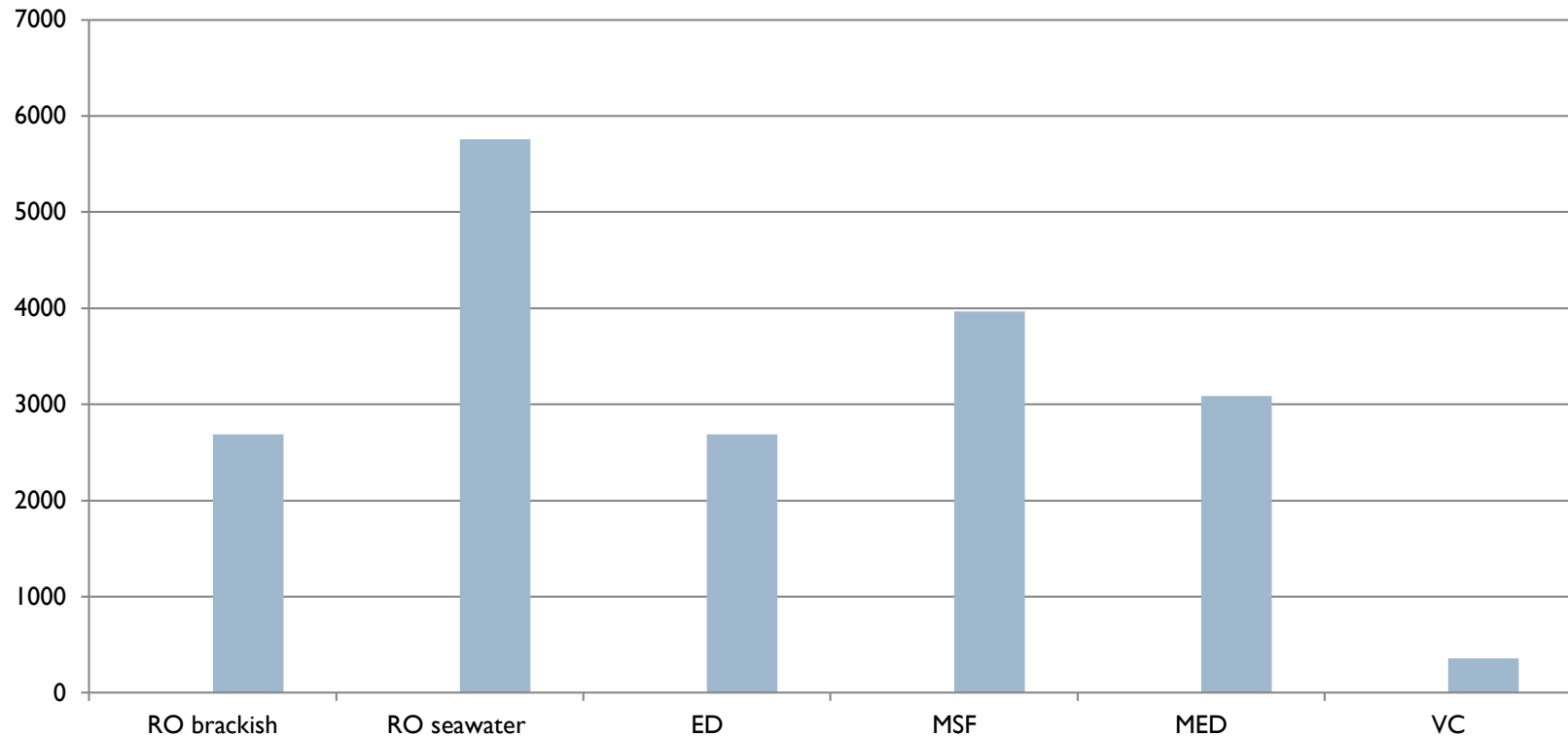


Selection of the Optimal Desalination System

Desalination Method	Description
Reverse Osmosis (RO)	Pressure driven process that currently is the most widely used method. Based on overcoming natural osmotic pressure gradient which is reversed by application of a hydraulic force across a semi-permeable membrane.
Forward Osmosis (FO)	Similar to RO, water transports across a membrane that is impermeable to salt. FO, however, utilizes an osmotic pressure gradient induced by a draw solution.
Electrodialysis (ED)	Electrochemical process for the separation of ions across charges of membranes with the electro potential difference as the driving force.
Multi-Stage Flash (MSF)	Water heated through a series of stages, each with successively lower temperatures and pressures. Flashing or rapid boiling is caused by the reduced pressure. The flashed water is condensed and collected.
Multi-Effect Distillation (MED)	Consists of multiple stages (effects) and in each stage the water is heated by steam in tubes. Some water evaporates and (as steam) flows into the tubes of the next stage. Steam is condensed and collected.
Vapour Compression (VC)	Evaporation of water through heat from the compressed vapour which was obtained from distillation.



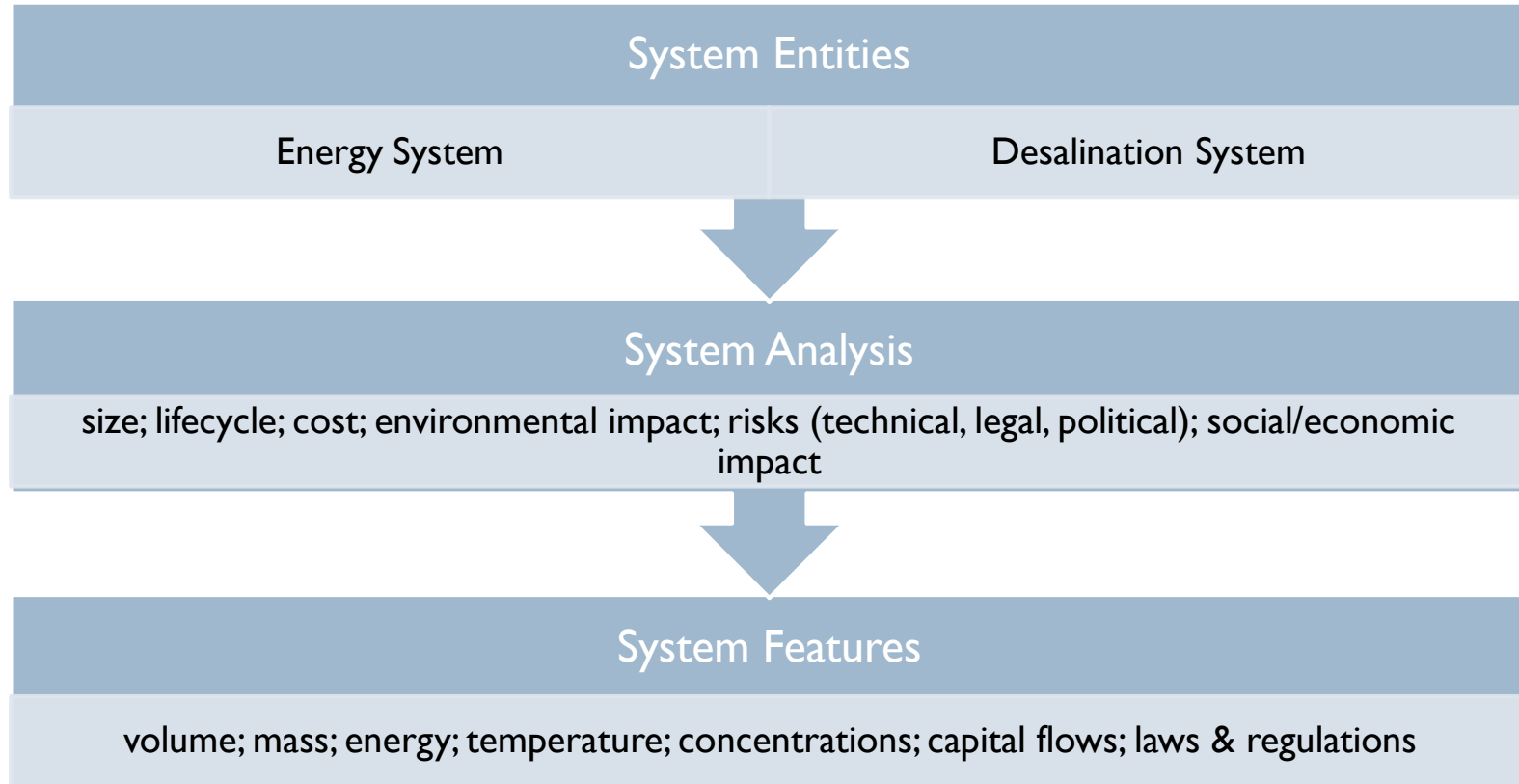
Desalination: Multi-Criteria Analysis



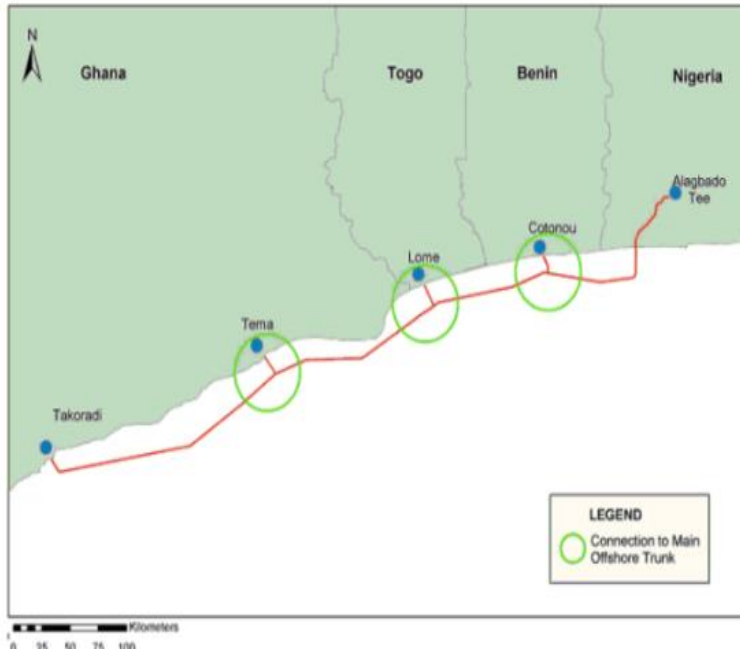
Reverse Osmosis of seawater is the optimal technology in this case. This conclusion is consistent with similar multi-criteria analyses applied to cogeneration of water and power. For example, a study to determine the optimal desalination method for an energy-rich Arabian Gulf region concluded that reverse osmosis was most appropriate, despite MSF systems being most commonly built in the region.



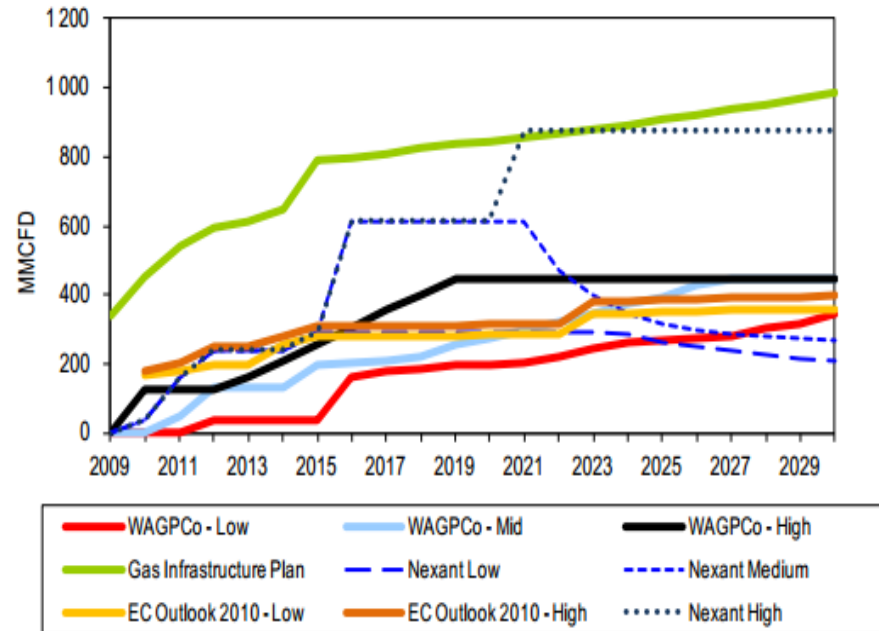
System Features



Energy System Analysis



West Africa Gas Pipeline connects to the Tema/Accra urban area.



Forecasts for NG supply vs. demand show if planned gas infrastructure investments are made, the supply will be sufficient.

Plant Concept

- Generation capacity will have to expand 1,500MW within 5-years to meet demand requirements.
- An available gas supply of 5.6 million m³/d is assumed feasible in the near term. $Q_{in} = 5.6 \text{ million cubic meters/d} = 53.0 \text{ kg/s}$

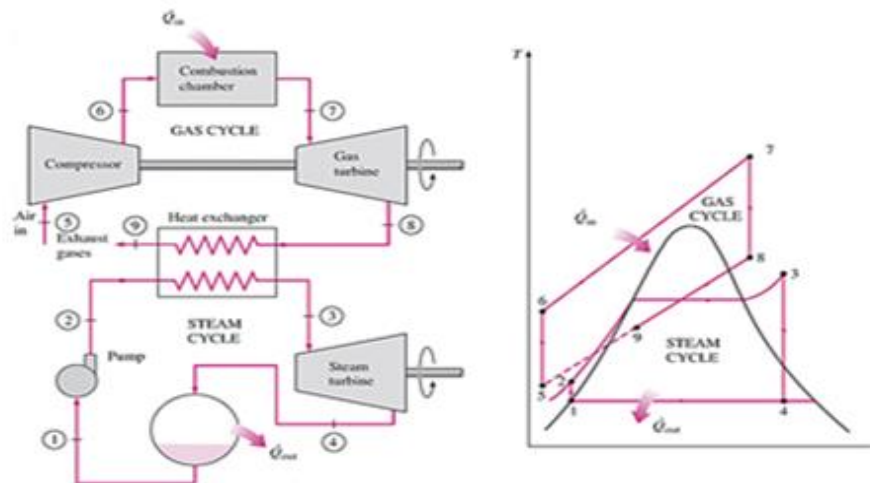
Combined Cycle Model Assumptions

Parameter	Value
Air-fuel ratio	42
Mass flow rate (fuel) [kg/s]	53.0
Pressure Ratio	18.5
Compressor Inlet Gas temperature [K]	300
Turbine Inlet Gas temperature [K]	1644
Turbine efficiency [%]	90
Compressor efficiency [%]	90
Relative humidity [%]	60

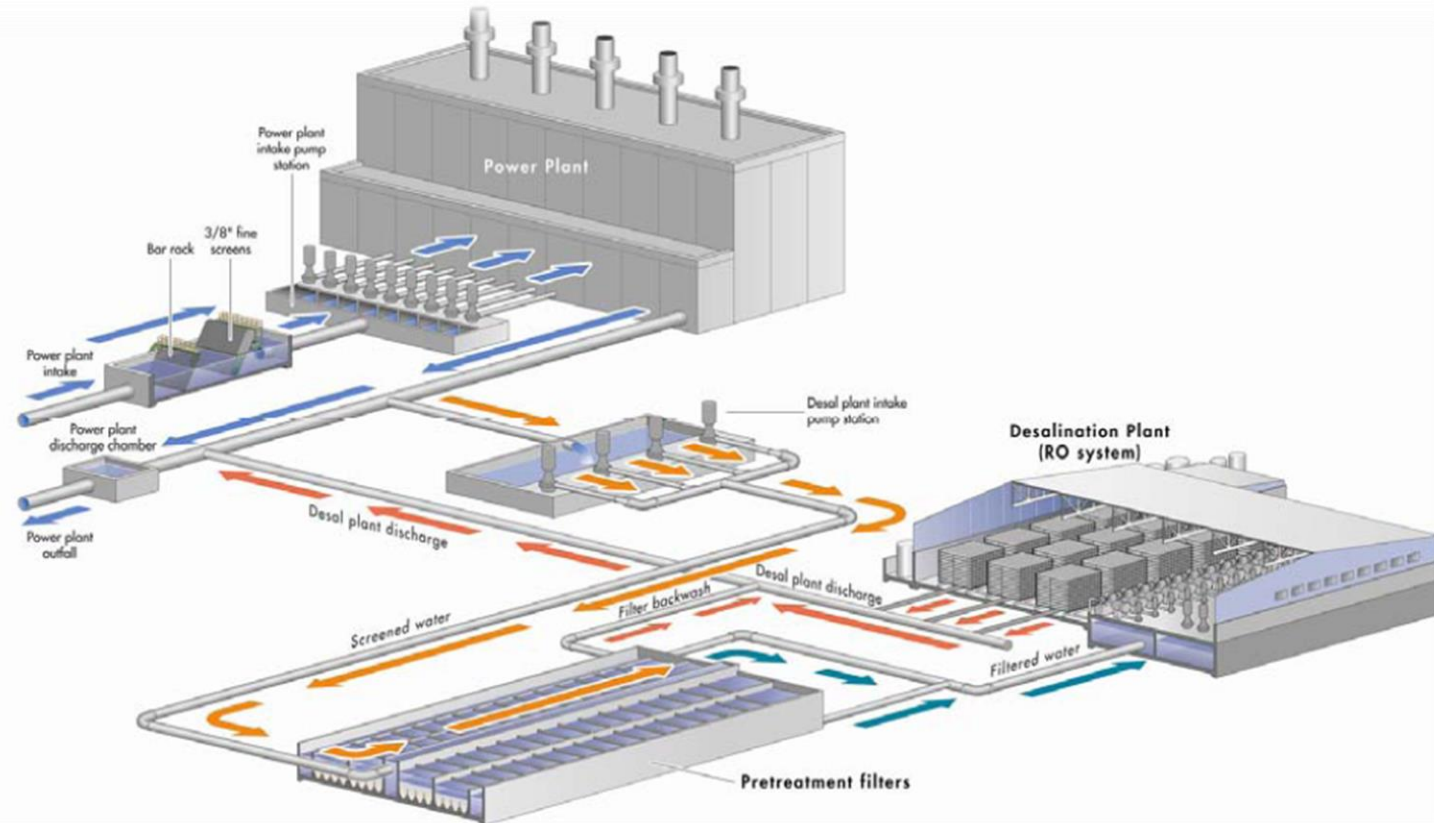


Plant Concept

Power plant operating parameters	
Cooling media	Seawater
Water quantity (60-100x exhaust steam flow)	1,413,542.37 m ³ /day (58 x steam flow)
Fuel LHV	47 MJ/kg
Gas supply rate	53 kg/s
Efficiency	59%
Net output	1,426.11 MW

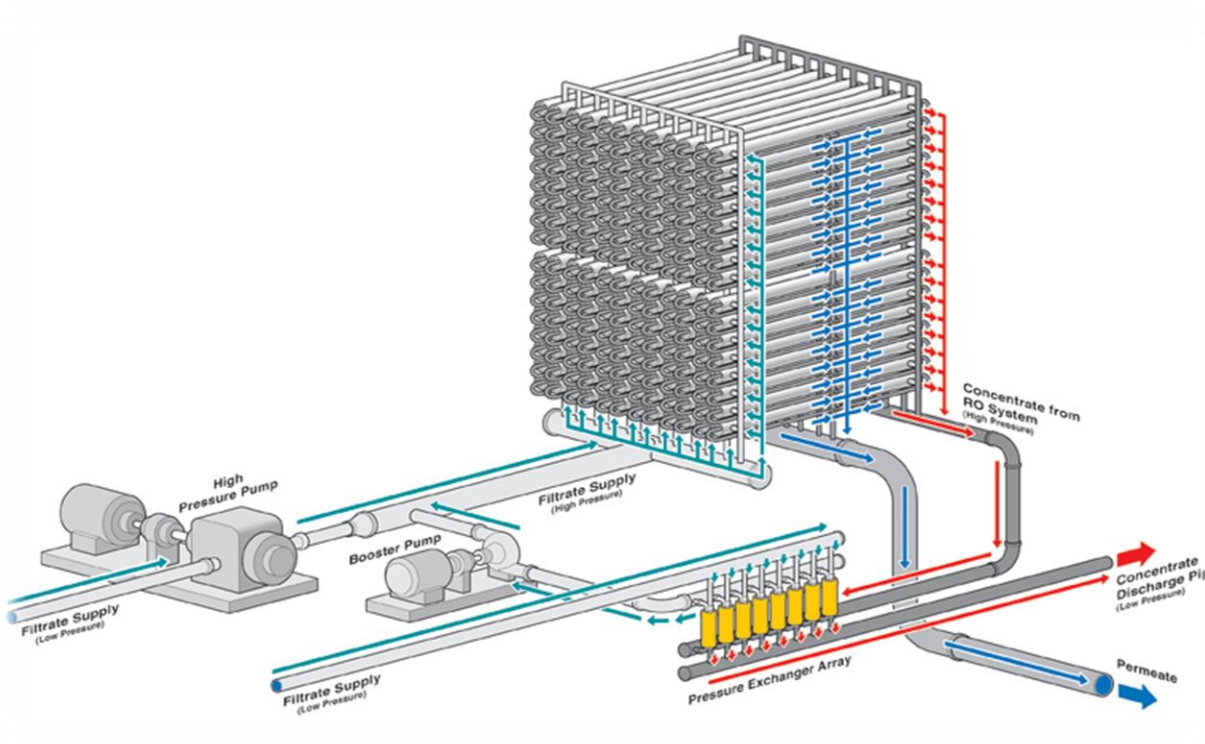


Combined Water Intake



Combined water intake system.

Reverse Osmosis



RO Pressure vessels arranged in trains of 100-200.

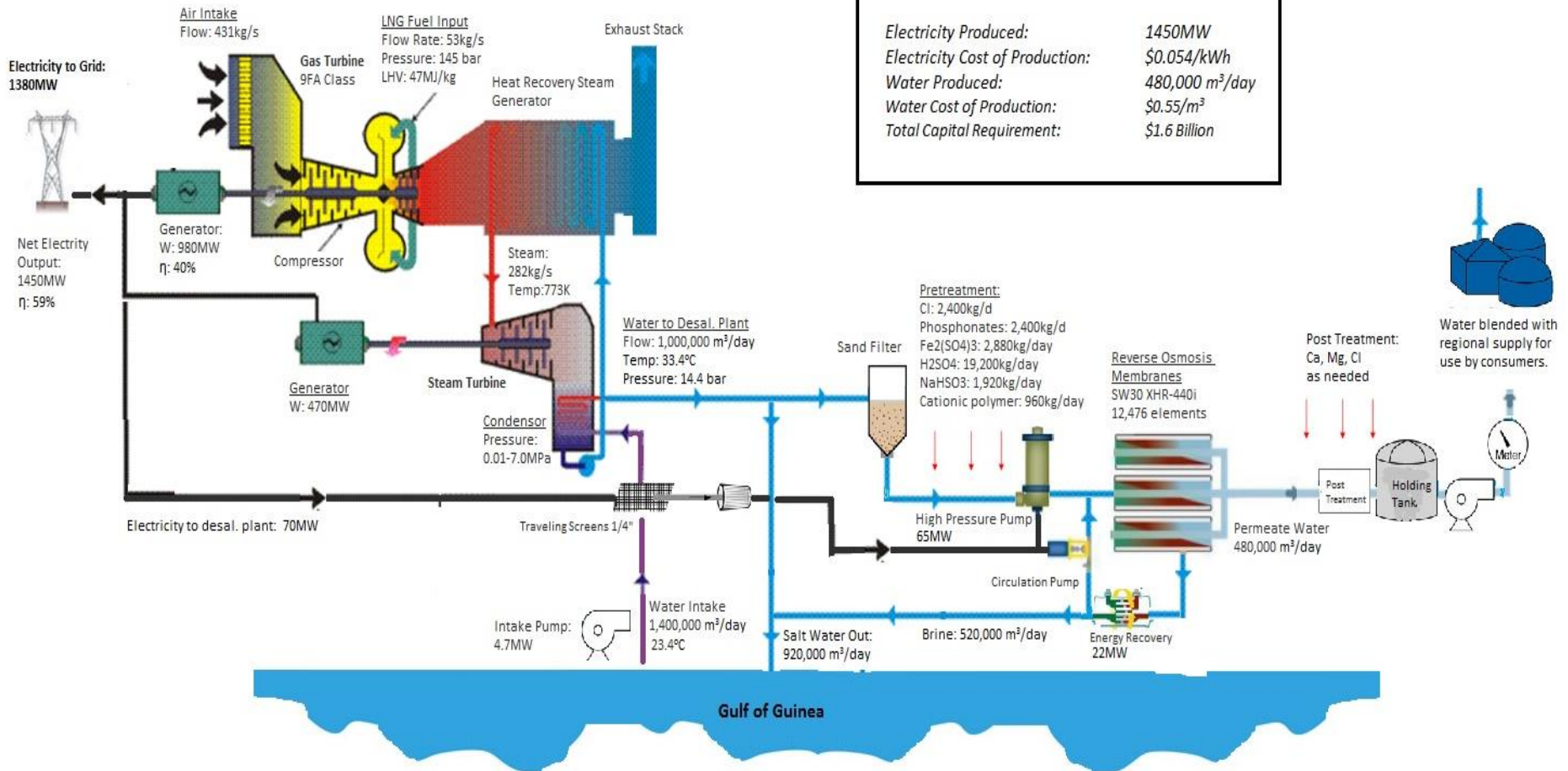
Process Flow Diagram

Ambient Conditions

Temp: 26.8°C
Pressure: 1.0 bar
Humidity: 60%

Cogeneration Plant: Key Figures

Electricity Produced: 1450MW
Electricity Cost of Production: \$0.054/kWh
Water Produced: 480,000 m³/day
Water Cost of Production: \$0.55/m³
Total Capital Requirement: \$1.6 Billion



RO Plant Parameters

Qf (m ³ /day)	1,000,000
energy for high pressure pump	4.35 kWh/m ³
energy of product pump	Na
energy recovery (kWh/m ³)	1.094
Net energy consumption (kWh/m ³)	3.25
production capability (m ³ /day)	480,000
raw water supply (m ³ /day)	1,000,000
recovery %	48%
Membrane	Dow SW 30XHR-440i



Plant in Ashkelon, Israel produces more than 100 million m³ of water/year and includes its own gas turbine power station to provide the facility with a dedicated electrical supply.

Production Cost of Water

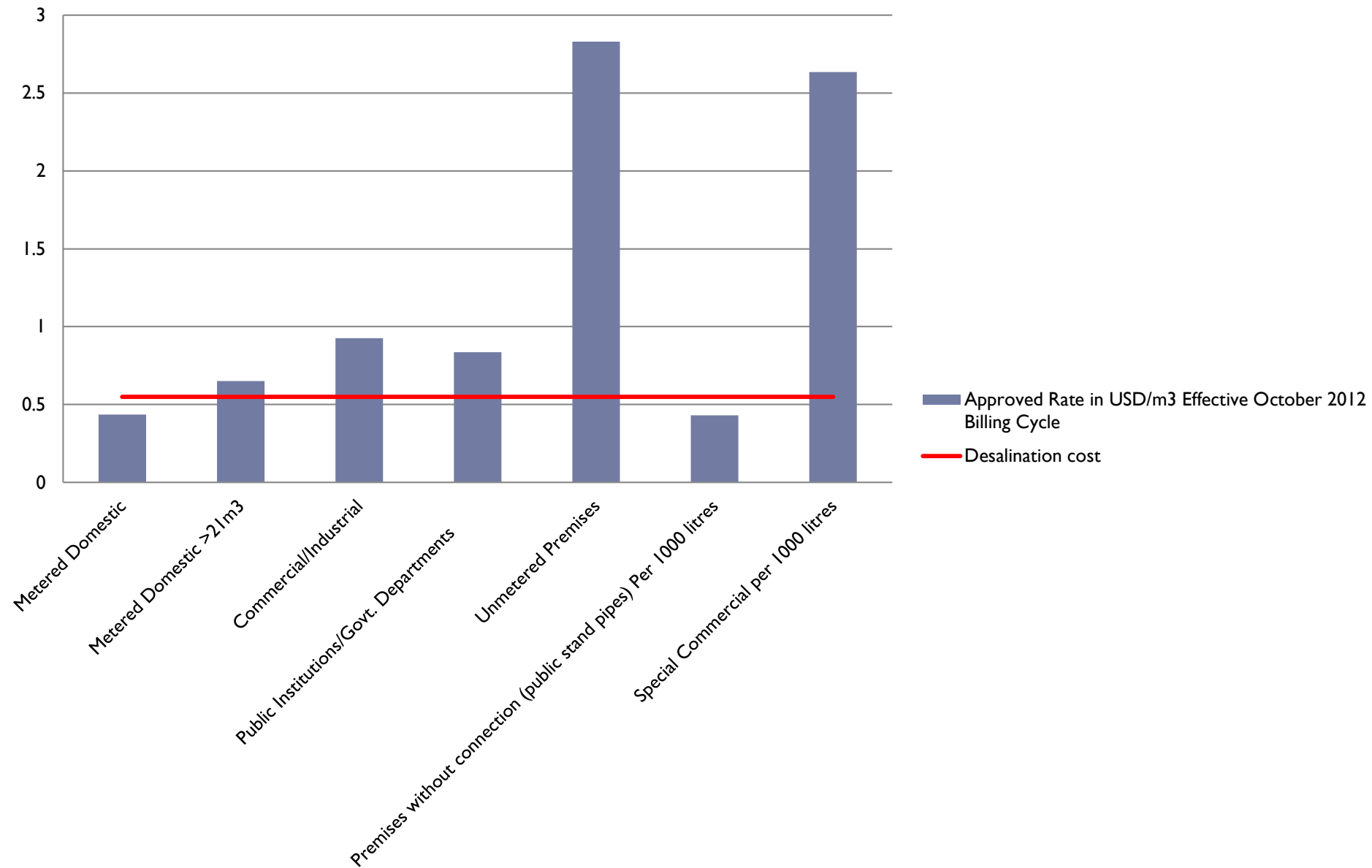
Capital Cost Estimate	Cost \$USD
Site preparation Cprep	3,000,000
Intake	0 (included in gas plant cost)
Pretreatment	24,600,000
RO System Equipment	111,000,000
Post-Treatment	6,200,000
Concentrate Disposal	0 (included in gas plant cost)
Residuals Management	4,600,000
Electrical System	7,700,000
Product Storage & Distribution	6,000,000
Buildings	15,400,000
Start-up Costs	6,200,000
Project engineering services	33,900,000
Administration, contracting, management	7,700,000
Environmental permitting	0 (included in gas plant cost)
Legal services	0 (included in gas plant cost)
Project financing cost	37,000,000
Contingency	20,000,000
TOTAL CAPITAL COSTS	283,000,000
Unit Cost \$/m3	\$590.00

Operation Cost Estimate	Cost \$USD/year
Operation & Maintenance	1,826,400
Replacement Parts & Materials	14,268,000
Replacement Membranes	6,086,400
Power	30,747,600
Chemicals	14,500,800
TOTAL	\$28,535,160

The total unit cost of water is the sum of operation and maintenance costs plus the amortized capital cost (over 25 years at 5% interest).

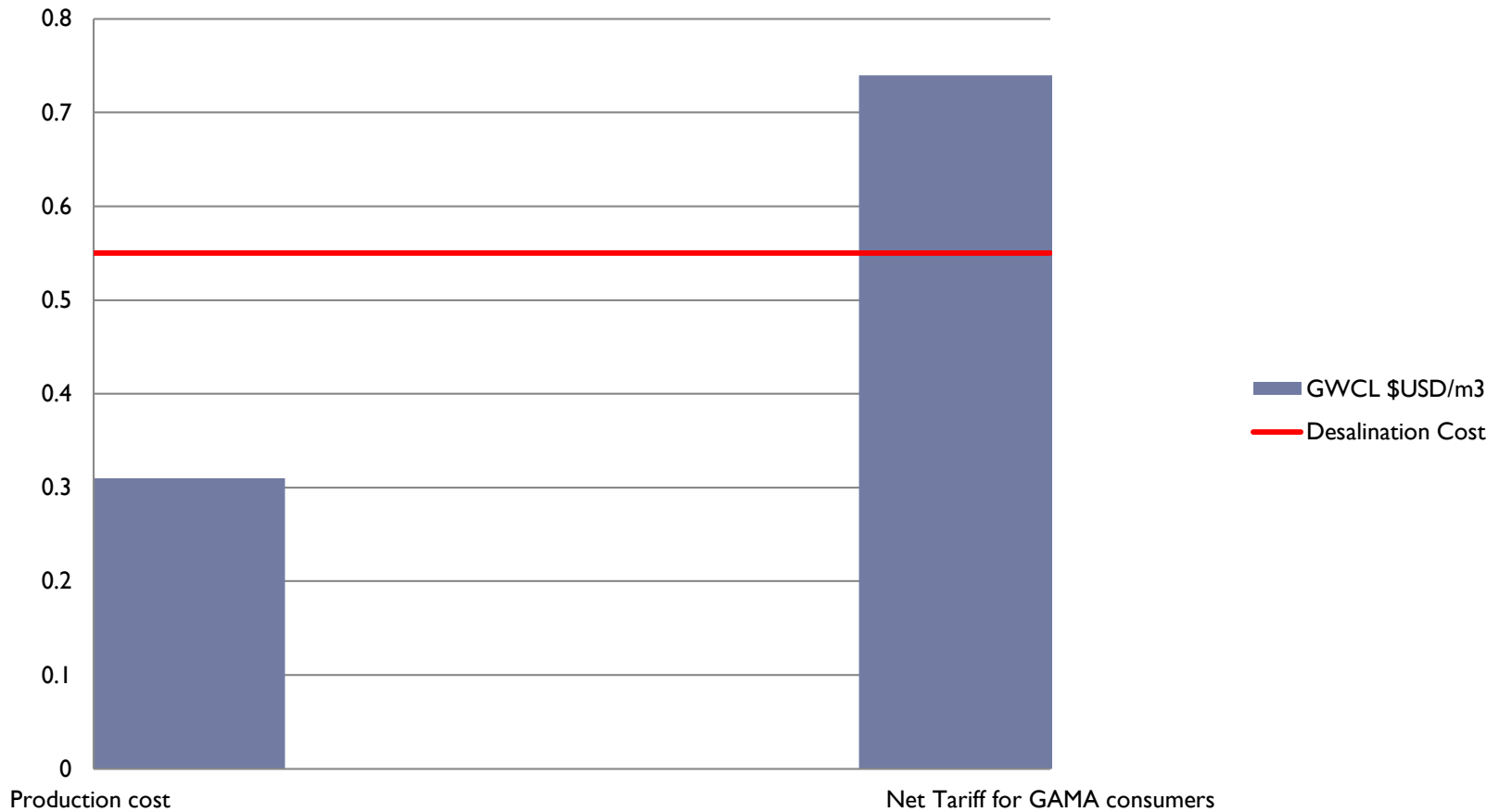
$$\text{Unit Cost} = \$0.55/\text{m}^3$$

GWCL Water Tariffs \$USD/m³ comparted with desalination cost



data source: GWCL, 2012

Cost of GWCL water and desalination



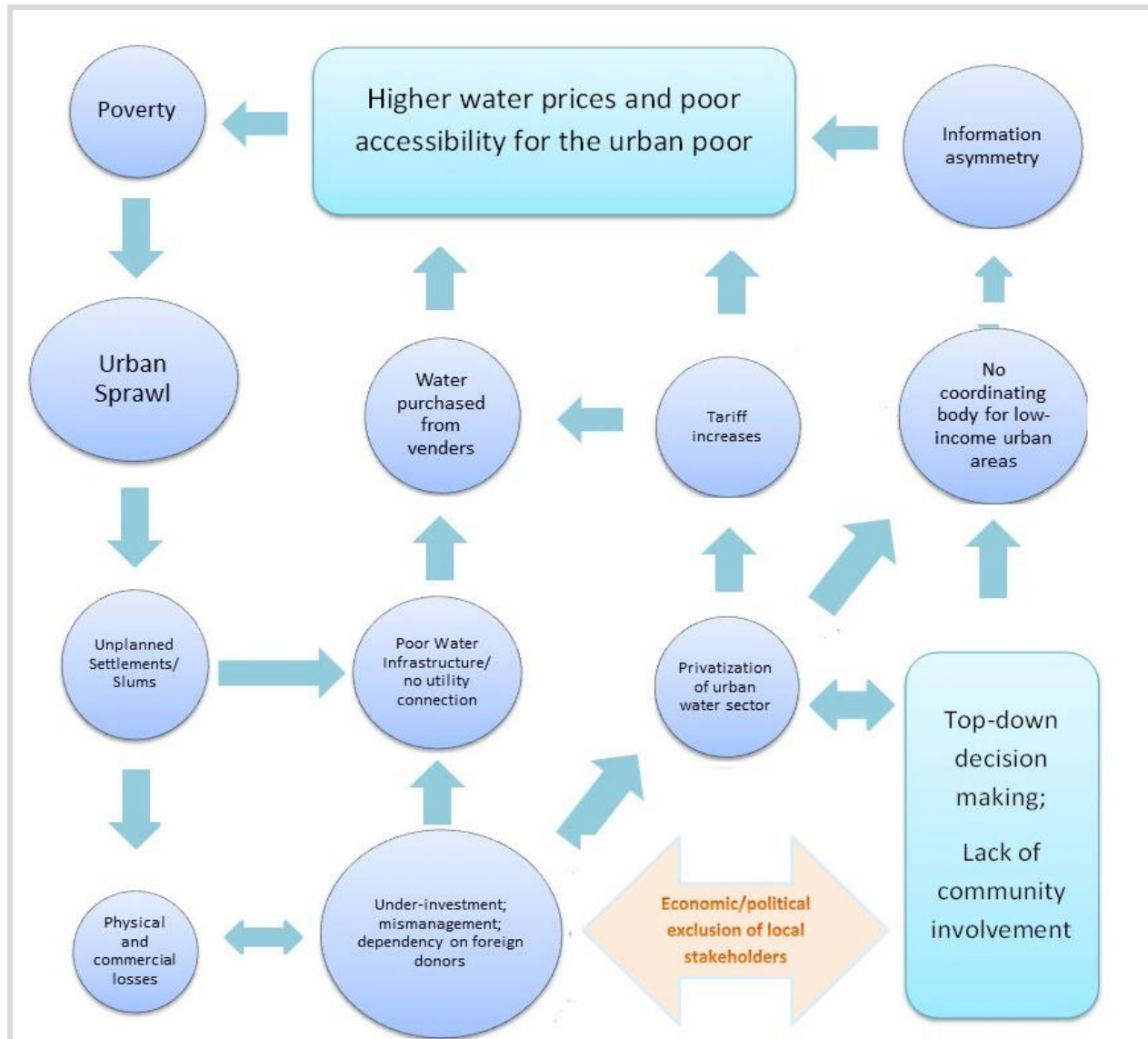
The benefits of water security and added supply capacity that come with desalination perhaps compensate for the higher production cost.

Affordability and Access

- 90% of high income areas and only 16% of low income zones have a water system connection (in house or neighbourhood) (Ainuson, 2010); 50% rely on vendors for potable water supply.
- Most lower-income residents depend on intermediate suppliers at a cost up to 12 times that charged by the utility and spend considerable percentage of household income on water (SWITCH EU, 2011).
- Current water policy seems has focus on increasing system capacity (expanding water supply/treatment plants); there is relatively little attention or investment to other areas, particularly improving access and affordability for the urban poor.



Water Affordability and Access



Problem diagram for lack of accessibility and affordability of potable water in the urban area of Greater Accra.

Top down decision making and lack of community involvement as described by (Fuest & Haffner, 2010) is seen as an ongoing management pattern that framed human decisions leading to the current problems.

A desalination plant does NOT solve this problem.

Preliminary Conclusions

- Desalination of seawater on a large scale is possible in the Greater Accra region and seawater could be an abundant, affordable supply source for most (though not all) consumers.
- The cost of desalination remains significantly higher than the main freshwater source, Lake Volta. Lake Volta will likely continue to be the main potable water source for the growing metropolis. However, desalination would add security of supply.
- If a large desalination plant is built, cogeneration with a natural gas plant could present cost savings and improved system performance.
- Current system upgrades, such as reducing water system losses and electrical transmission losses, should take priority over any supply expansions (though both will be needed within the next 10 years).
- Legal and institutional reforms are needed to allow the urban poor to benefit from any added water and energy infrastructure (outlined in Ch. 5);
- Desalination is not the answer to the underlying issues in the urban water sector, which are largely organizational and social, rather than technical problems.



Questions & Discussion

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